Predictive Modeling of Serine Protease Substrate Catalysis

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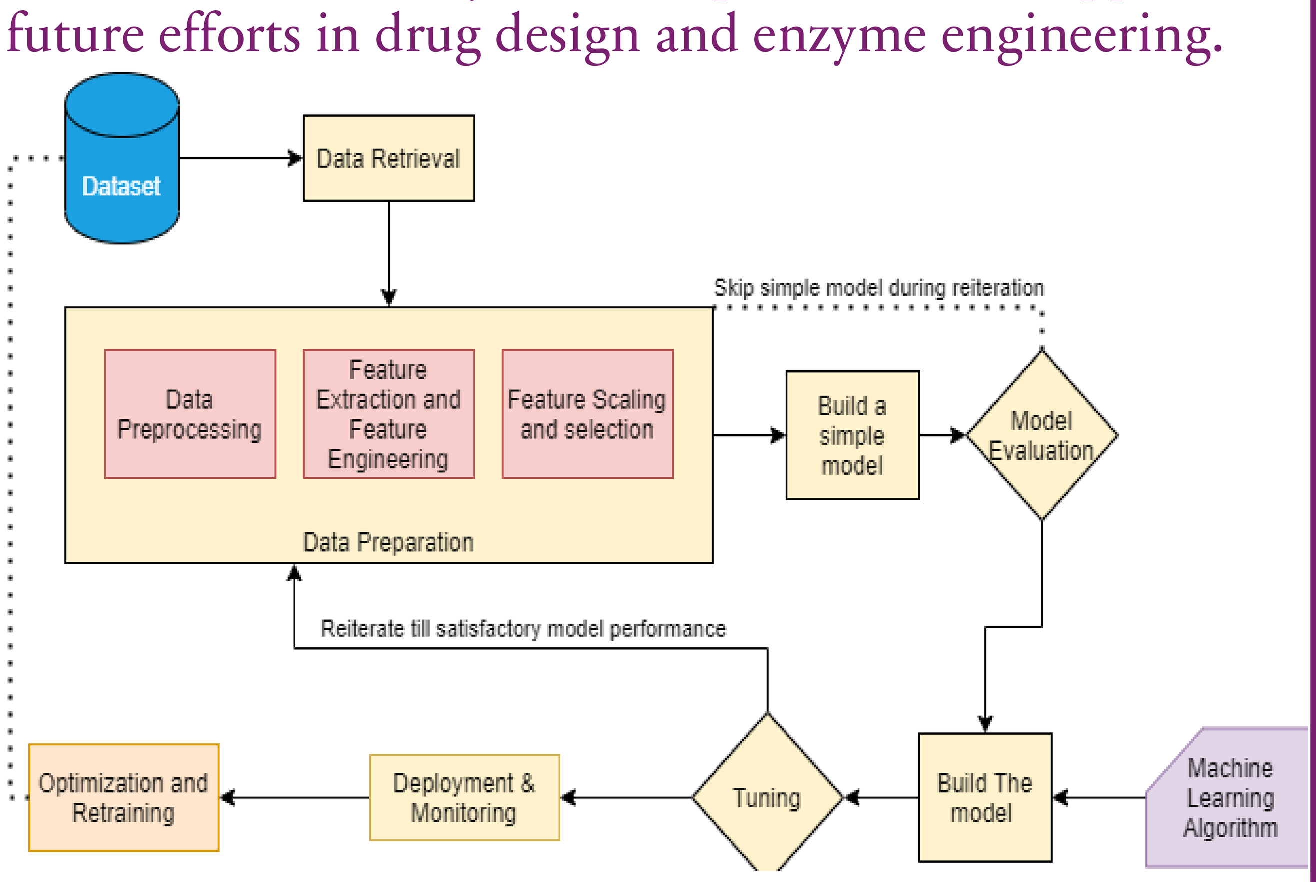
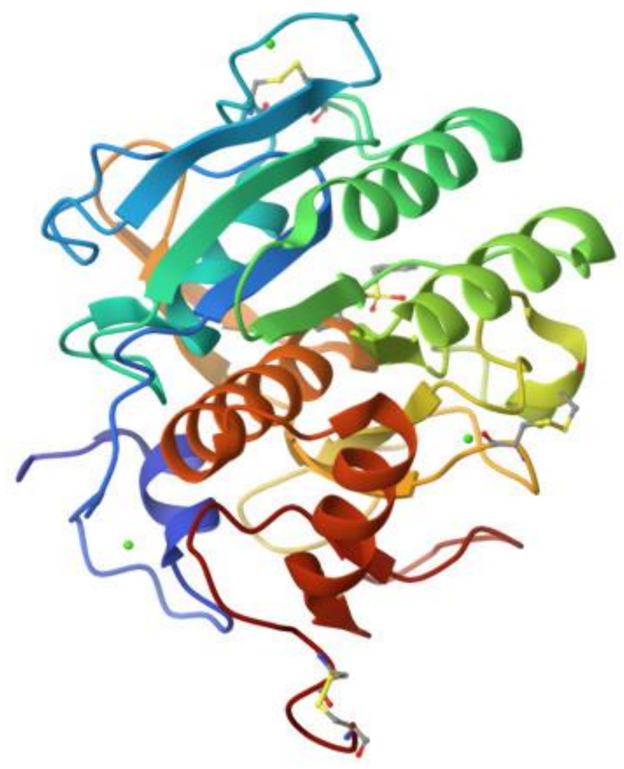
Background

Trypsin-like serine proteases play key roles in digestion, coagulation, and immune regulation. This study focuses on **1SH7**, a well-characterized trypsin-like protease, to investigate the molecular determinants of catalytic activity. We analyze key kinetic parameters: **kcat**, **Km**, and **kcat/Km,** in relation to substrate features such as charge, size, and binding pocket compatibility. By examining how active site geometry and substrate properties modulate catalysis, we aim to better understand efficiency in serine proteases and support future efforts in drug design and enzyme engineering.



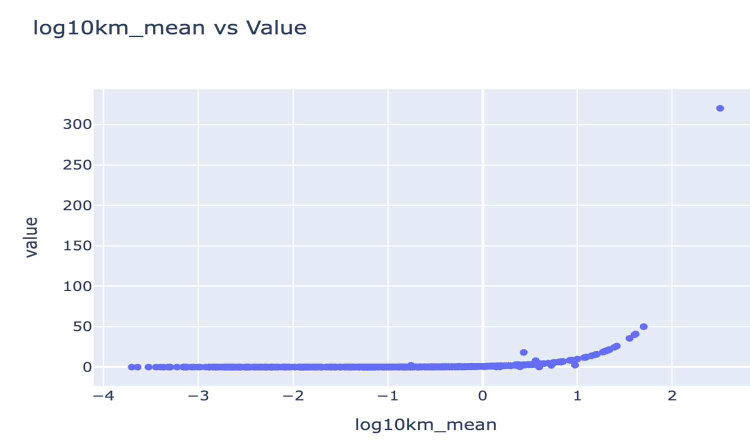


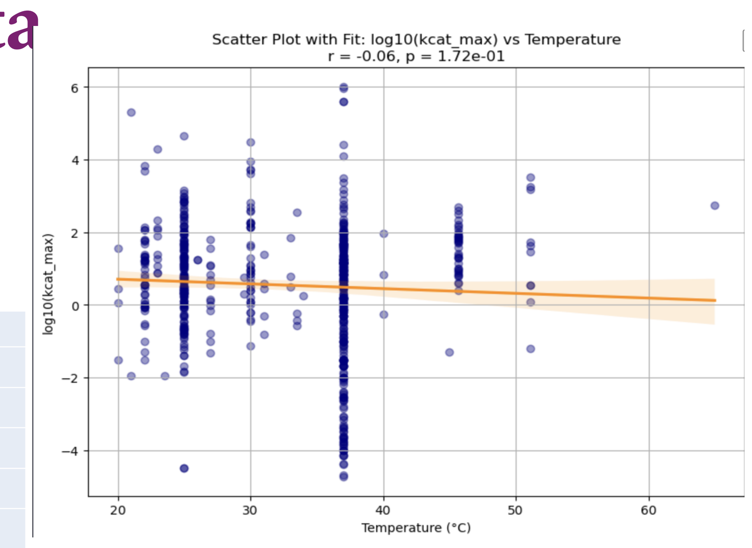
Methodology

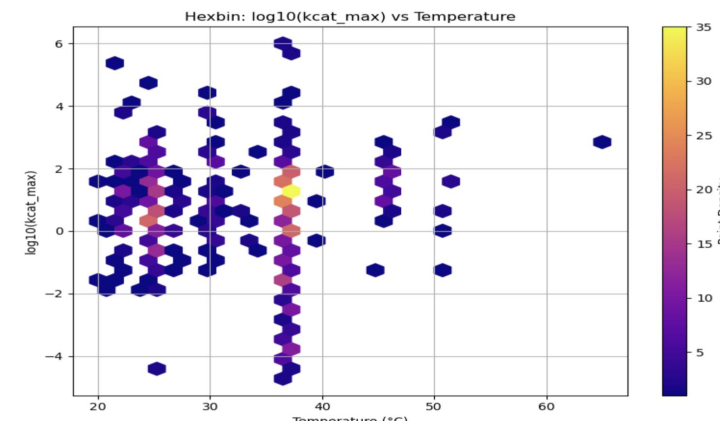
The Process

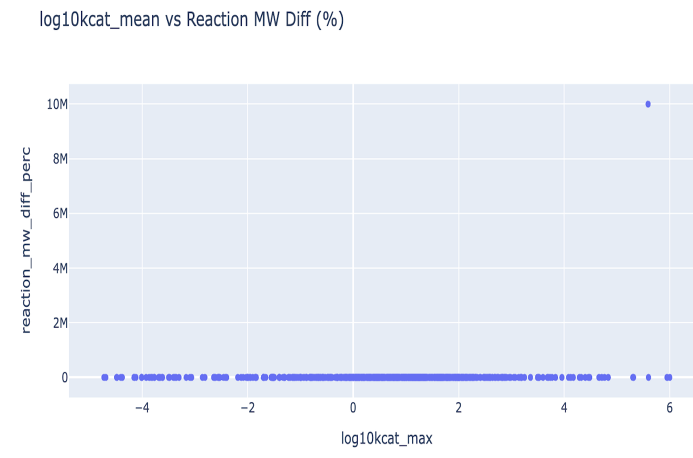
- **Data Pipeline**
  - PDB data → substrate SMILES → ligand preparation
  - Docking (AutoDock, DiffDock)
- **MD Simulation**
  - Ligand-enzyme complex
  - Parameterization of ligands
- **ML Modeling**
  - GNN
  - Input: 3D structures + SMILES
  - Transfer learning on pre-trained models

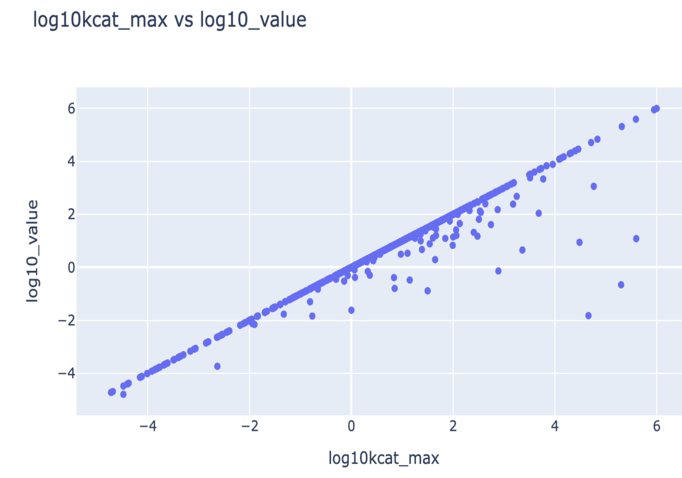
Interpreting the Data

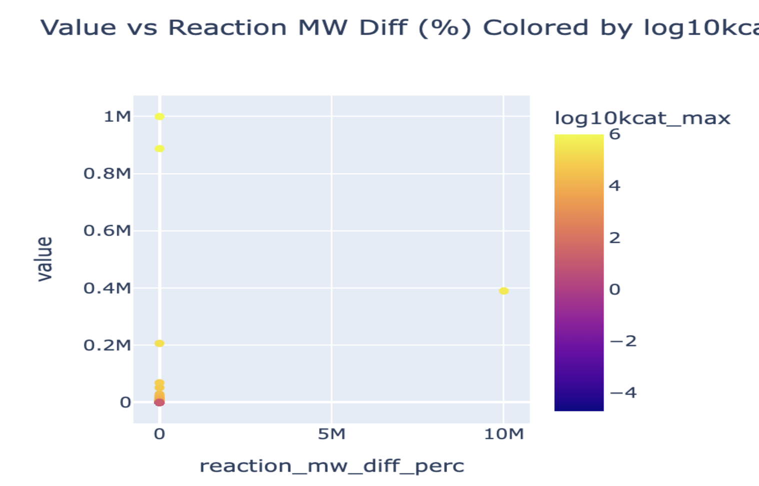


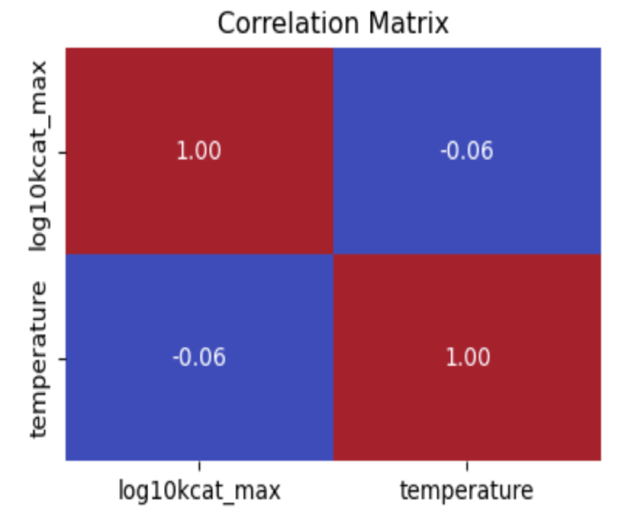


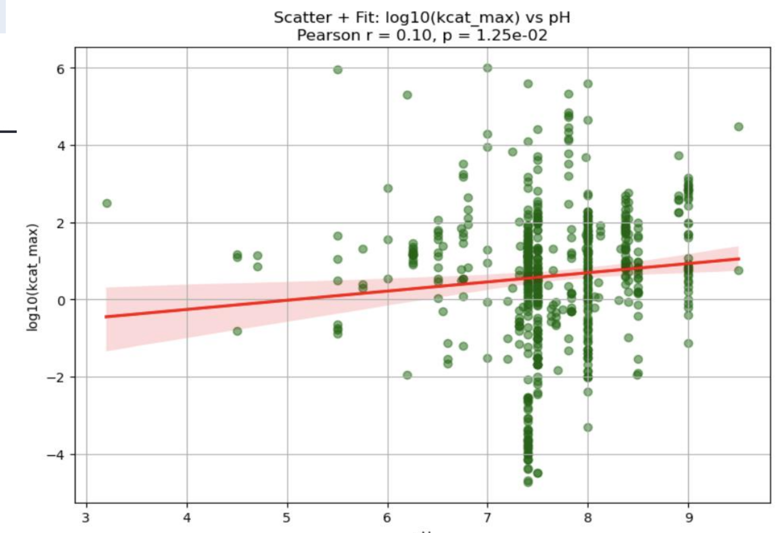




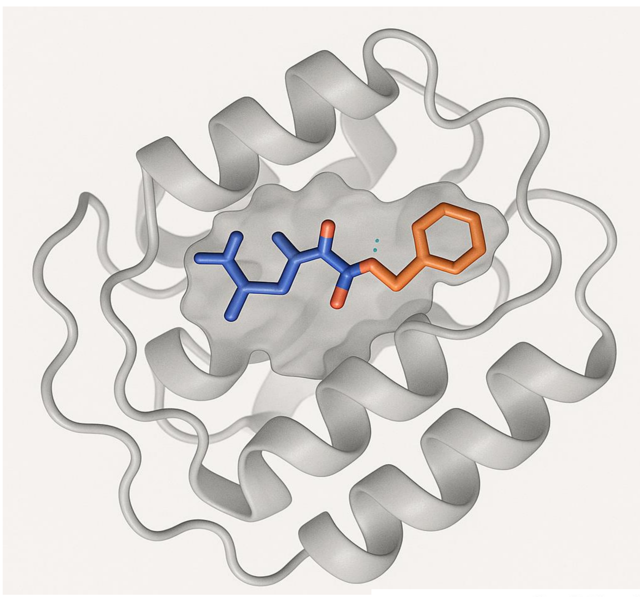


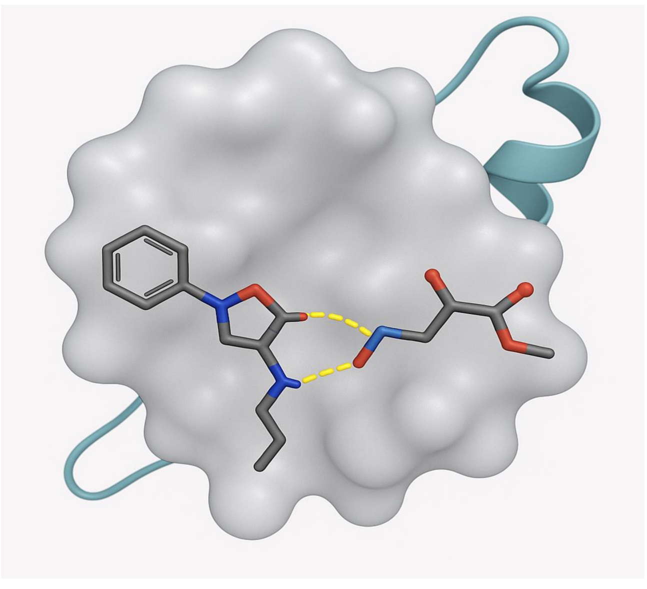


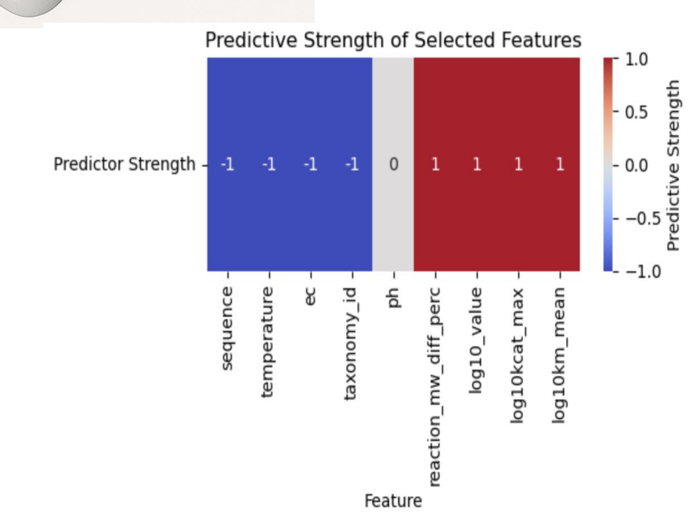




Complex Analysis







Prediction Model

Simple Logistic Regression – Univariate (Current)
Input: single feature (molecular weight only, LogP only)
Output: high or low catalytic efficiency

Graph Neural Network (Planned)
Input: graphs of ligand-enzyme complexes
Node (atoms) features: atom types, hybridization, partial charge
Edges (bonds) features: bond order proximity in docked position

Results

We combined ligand descriptors with catalytic efficiency data and filtered out missing or non-informative features. Catalytic efficiency (**kcat/Km**) was binarized into "high" vs. "low" classes. Only activators were included to focus on features that promote catalysis. Univariate logistic regression showed that **ligand polarity** was the strongest predictor (~68% accuracy, AUC > 0.70), followed by **ligand charge**.

Conclusion

We developed an early workflow that, building on these insights, will employ multivariate and graph neural network models to capture complex structural and spatial determinants of catalysis. Expanding datasets to include diverse substrates and inhibitors aims to enhance model robustness and guide scalable enzyme design in synthetic environments.

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